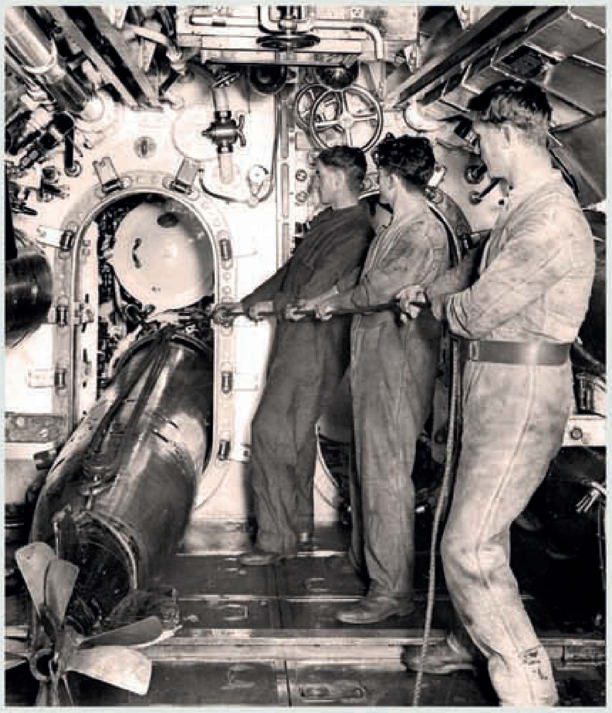
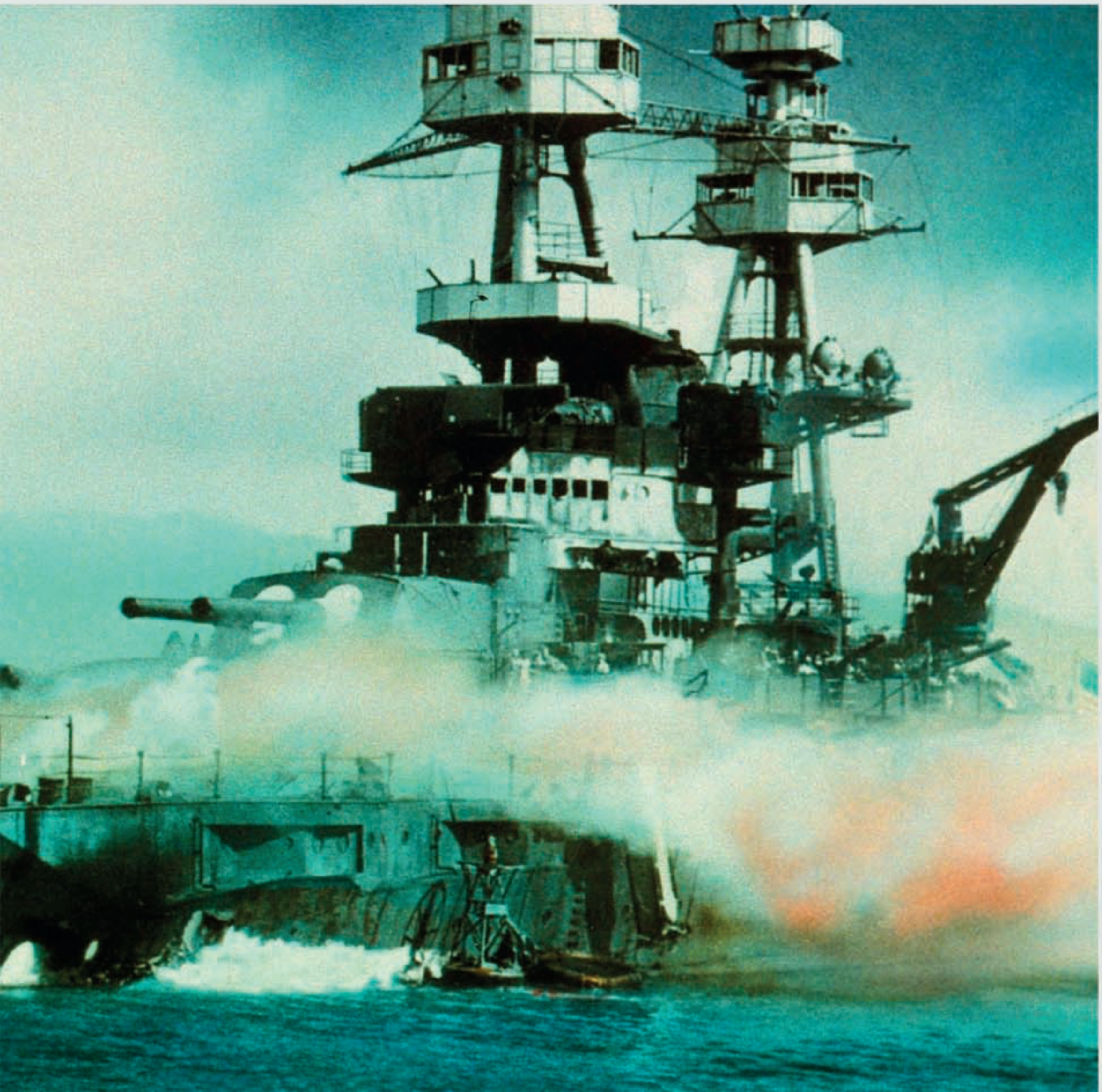
at the start of the war, until the technique of degaussing ship's hulls (reducing their magnetism) was introduced.

Meanwhile, U-boats, coordinated by radio and with improved range, were even more destructive to merchant shipping than they had been in World War I. A combination of tactical and technological innovations were employed to meet this threat: these included improved radar on convoy escort ships, and high-frequency radio direction-finding equipment ("huff-duff").

When gun battles between large surface warships did occur, victory usually went to the side with superior radar and night-fighting equipment, as in the fighting between Japanese and American fleets around Guadalcanal, in 1942, or between the British and Italians in the Mediterranean. Eventual naval superiority allowed the Allies to mount numerous amphibious landings: the largest of these, at Normandy in June 1944, was supported by over 1,200 warships.



◂ **LOADING A TORPEDO**
The cramped conditions on World War II submarines made loading torpedoes in the torpedo tubes extremely difficult. On German U-boats – as with British submarines – the torpedo room also served as living accommodation.



◂ **PEARL HARBOR**
The USS *Nevada* was among the 18 American ships sunk or damaged in Japan's surprise attack on the US Pacific fleet on 7 December 1941, at Pearl Harbor. The Japanese force consisted of 353 fighter, bomber, and torpedo aircraft launched from six aircraft carriers, supported by midget submarines.

KEY **FIGURE**

ADMIRAL ISOROKU YAMAMOTO
1884–1943

Having fought as an ensign at the Battle of Tsushima in 1904, Yamamoto rose to be commander-in-chief of the Japanese Combined Fleet by 1939. An advocate of naval air power, he planned the attack on Pearl Harbor that started the Pacific War in 1941.

▲ Yamamoto died when his transport aircraft was ambushed and shot down by US fighters.